

B3: Quantum, Atomic & Molecular Physics — Model Solutions, Trinity 2015

Question 1 — Central field, LS coupling, 5s5l of Sr

Problem. Define central field approximation, configurations, terms, levels. Derive the Landé interval rule. Identify l in Sr 5s5l from three given level energies and assign L, S, J to all four levels. From six observed transition energies, identify l' in 5s5l' and assign L, S, J on both ends.

(a) Central field approximation, configurations, terms, levels

Replace the full electron–electron repulsion by a spherically symmetric one-body potential $U(r)$ (the spherical average of the nuclear attraction plus the mean field of the other electrons), and treat the residual

$$H_{\text{res}} = \sum_{i < j} \frac{e^2}{4\pi\epsilon_0 r_{ij}} - \sum_i [U(r_i) + Ze^2/(4\pi\epsilon_0 r_i)]$$

as a perturbation.

In $U(r)$ each electron has good (n, l, m_l, m_s) and energies E_{nl} , so the zeroth-order eigenstate is fully specified by populations $(nl)^q$ — the *configuration*.

Residual Coulomb H_{res} commutes with total $\mathbf{L}$ and $\mathbf{S}$ but not with individual $\mathbf{l}_i, \mathbf{s}_i$:

$$\text{configuration} \longrightarrow \text{terms } {}^{2S+1}L, \quad \text{splitting} \sim 10^4 \text{ cm}^{-1}.$$

Spin–orbit $H_{\text{SO}} = \sum_i \xi(r_i) \mathbf{l}_i \cdot \mathbf{s}_i$ commutes only with total $\mathbf{J} = \mathbf{L} + \mathbf{S}$:

$$\text{term} \longrightarrow \text{levels } {}^{2S+1}L_J, \quad J = |L - S|, \dots, L + S.$$

(b) LS coupling and the Landé interval rule

LS coupling: $H_{\text{res}} \gg H_{\text{SO}}$, so L, S are good quantum numbers and H_{SO} acts within a fixed ${}^{2S+1}L$ term.

Project $\sum_i \xi(r_i) \mathbf{l}_i \cdot \mathbf{s}_i$ onto the term: by the projection (Wigner–Eckart) theorem,

$$H_{\text{SO}}|_{LS} = A_{LS} \mathbf{L} \cdot \mathbf{S}. \tag{1}$$

Use $\mathbf{J} = \mathbf{L} + \mathbf{S}$:

$$\mathbf{L} \cdot \mathbf{S} = \frac{1}{2}(\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2).$$

Eigenvalue in level J :

$$E_J = \frac{\hbar^2}{2} A_{LS} [J(J+1) - L(L+1) - S(S+1)].$$

Adjacent-level interval:

$$E_J - E_{J-1} = \frac{\hbar^2}{2} A_{LS} [J(J+1) - (J-1)J].$$

Simplify:

$$\boxed{E_J - E_{J-1} = \hbar^2 A_{LS} J} \quad (2)$$

The spacing between adjacent levels of a fine-structure multiplet is proportional to the larger J .

(c) Identifying l and assigning L, S, J for $5s5l$

Configuration $5s5l$ has $s_1 = s_2 = \frac{1}{2}$, $l_1 = 0$, $l_2 = l$, hence $L = l$, $S = 0$ or 1 :

four levels: 1L_l , $^3L_{l-1}$, 3L_l , $^3L_{l+1}$.

The triplet is the only multiplet showing fine structure; the singlet is the isolated fourth level.

Triplet intervals from the data:

$$\Delta_1 = 35022 - 35007 = 15 \text{ cm}^{-1}, \quad \Delta_2 = 35045 - 35022 = 23 \text{ cm}^{-1}.$$

Apply (2) with $J = L, L + 1$ for the two adjacent gaps:

$$\frac{\Delta_2}{\Delta_1} = \frac{L+1}{L}.$$

Insert numbers:

$$\frac{23}{15} \simeq \frac{3}{2} \Rightarrow L = 2.$$

Hence $l = 2$ ($5s5d$), and the triplet is $^3D_{1,2,3}$.

Assignments:

35 007 cm^{-1} :	3D_1 :	$L = 2, S = 1, J = 1,$
35 022 cm^{-1} :	3D_2 :	$L = 2, S = 1, J = 2,$
35 045 cm^{-1} :	3D_3 :	$L = 2, S = 1, J = 3,$
fourth level :	1D_2 :	$L = 2, S = 0, J = 2.$

(d) Identifying l' in $5s5l'$

Six transitions $\Rightarrow$ a triplet-triplet multiplet with $\Delta L = \pm 1, \Delta S = 0, \Delta J = 0, \pm 1$ (no $0 \rightarrow 0$).

Trial $l' = 1$ ($5s5p$, $^3P_{0,1,2}$): selection rules give

$$^3D_1 \rightarrow ^3P_{0,1,2} (3), \quad ^3D_2 \rightarrow ^3P_{1,2} (2), \quad ^3D_3 \rightarrow ^3P_2 (1).$$

Total: six. Consistent.

Cluster the data by lower level. Three transitions to the same lower level differ by the upper- 3D intervals 15, 23 cm^{-1} :

$$20\,108, 20\,123, 20\,146 : \quad \text{intervals } 15, 23 \Rightarrow \text{common lower } ^3P_2.$$

Two transitions to the same lower level differ by the upper- $^3D_1 \rightarrow ^3D_2$ interval 15 cm^{-1} :

$$20\,503, 20\,518 : \quad \text{interval } 15 \Rightarrow \text{common lower } ^3P_1.$$

Single transition:

$$20\,689 : \quad \text{from } ^3D_1 \text{ only} \Rightarrow \text{lower } ^3P_0.$$

Lower-level energies:

$$E(^3P_2) = 35\,007 - 20\,108 = 14\,899 \text{ cm}^{-1},$$

$$E(^3\text{P}_1) = 35\,007 - 20\,503 = 14\,504 \text{ cm}^{-1},$$

$$E(^3\text{P}_0) = 35\,007 - 20\,689 = 14\,318 \text{ cm}^{-1}.$$

Landé check ($J = 1, 2$):

$$\frac{E_2 - E_1}{E_1 - E_0} = \frac{14\,899 - 14\,504}{14\,504 - 14\,318} = \frac{395}{186} \simeq 2,$$

matching $(L+1)/L = 2/1$ for $L = 1, S = 1$. Confirms ^3P .

$$l' = 1 \text{ (5s5p)}, \quad \text{lower } ^3\text{P}_{0,1,2} : L = 1, S = 1, J = 0, 1, 2.$$

Per-line assignments (upper $\rightarrow$ lower):

$$\begin{aligned} 20\,108 \text{ cm}^{-1} : & \quad ^3\text{D}_1 (L, S, J = 2, 1, 1) \rightarrow ^3\text{P}_2 (1, 1, 2), \\ 20\,123 \text{ cm}^{-1} : & \quad ^3\text{D}_2 (2, 1, 2) \rightarrow ^3\text{P}_2 (1, 1, 2), \\ 20\,146 \text{ cm}^{-1} : & \quad ^3\text{D}_3 (2, 1, 3) \rightarrow ^3\text{P}_2 (1, 1, 2), \\ 20\,503 \text{ cm}^{-1} : & \quad ^3\text{D}_1 (2, 1, 1) \rightarrow ^3\text{P}_1 (1, 1, 1), \\ 20\,518 \text{ cm}^{-1} : & \quad ^3\text{D}_2 (2, 1, 2) \rightarrow ^3\text{P}_1 (1, 1, 1), \\ 20\,689 \text{ cm}^{-1} : & \quad ^3\text{D}_1 (2, 1, 1) \rightarrow ^3\text{P}_0 (1, 1, 0). \end{aligned}$$

Question 2 — Hyperfine structure in a magnetic field, ^{39}K

Problem. Identify the three terms in the Hamiltonian and define weak/strong field. Show $\Delta E_F = g_F \mu_B B_{\text{ext}} M_F$ in the weak limit and find g_F in terms of g_J . Compute g_J, F, g_F for ^{39}K 4s. Sketch the seven RF frequencies between hyperfine states and indicate which are visible along $\mathbf{B}$.

(a) Physical origin and field regimes

$A_J \mathbf{I} \cdot \mathbf{J}$: hyperfine interaction. The nuclear magnetic moment $\boldsymbol{\mu}_I = g_I \mu_N \mathbf{I} / \hbar$ couples to the magnetic field at the nucleus produced by the electrons; projection within a level gives $H_{\text{hf}} = A_J \mathbf{I} \cdot \mathbf{J}$ (Fermi contact for s -electrons; orbital + dipolar for $l \neq 0$).

$g_J \mu_B \mathbf{J} \cdot \mathbf{B}_{\text{ext}}$: Zeeman energy of the electronic magnetic moment $\boldsymbol{\mu}_J = -g_J \mu_B \mathbf{J} / \hbar$ in the external field.

$-g_I \mu_N \mathbf{I} \cdot \mathbf{B}_{\text{ext}}$: Zeeman energy of the nuclear moment in the external field; smaller than the electronic Zeeman by $\mu_N / \mu_B \sim 10^{-3}$.

Compare the electronic Zeeman $g_J \mu_B B_{\text{ext}}$ with the hyperfine spacing $\sim A_J |\mathbf{F}|$:

$$\text{weak: } g_J \mu_B B_{\text{ext}} \ll A_J F \Rightarrow \mathbf{F} = \mathbf{I} + \mathbf{J} \text{ good,}$$

$$\text{strong: } g_J \mu_B B_{\text{ext}} \gg A_J F \Rightarrow M_I, M_J \text{ good (Paschen–Back of hfs).}$$

(b) Weak-field shift and g_F

Couple $\mathbf{F} = \mathbf{I} + \mathbf{J}$; $|FM_F\rangle$ diagonalises $A_J \mathbf{I} \cdot \mathbf{J}$. Treat the Zeeman term as a perturbation, dropping the small nuclear Zeeman. First-order shift:

$$\Delta E_F = \langle FM_F | g_J \mu_B \mathbf{J} \cdot \mathbf{B}_{\text{ext}} | FM_F \rangle.$$

Project $\mathbf{J}$ onto $\mathbf{F}$ (Wigner–Eckart within a fixed F):

$$\mathbf{J} \rightarrow \frac{\langle \mathbf{J} \cdot \mathbf{F} \rangle}{F(F+1)\hbar^2} \mathbf{F}.$$

Evaluate $\mathbf{J} \cdot \mathbf{F} = \frac{1}{2}(\mathbf{F}^2 + \mathbf{J}^2 - \mathbf{I}^2)$:

$$\langle \mathbf{J} \cdot \mathbf{F} \rangle = \frac{\hbar^2}{2} [F(F+1) + J(J+1) - I(I+1)].$$

Take $B_{\text{ext}} = B_{\text{ext}} \hat{\mathbf{z}}$, $\langle F_z \rangle = \hbar M_F$:

$$\Delta E_F = g_J \mu_B B_{\text{ext}} M_F \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)}.$$

Identify g_F :

$$\Delta E_F = g_F \mu_B B_{\text{ext}} M_F, \quad g_F = g_J \frac{F(F+1) + J(J+1) - I(I+1)}{2F(F+1)}. \quad (3)$$

(c) ^{39}K 4s: g_J , F , g_F

Ground term 4s $^2\text{S}_{1/2}$: $L = 0$, $S = \frac{1}{2}$, $J = \frac{1}{2}$. Landé formula:

$$g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}.$$

Insert:

$$g_J = 1 + \frac{\frac{3}{4} + \frac{3}{4} - 0}{2 \cdot \frac{3}{4}} = 1 + 1.$$

$$g_J = 2.$$

Couple $J = \frac{1}{2}$ with $I = \frac{3}{2}$:

$$F = |J - I|, \dots, J + I = 1, 2.$$

Apply (3), $J(J+1) = \frac{3}{4}$, $I(I+1) = \frac{15}{4}$:

$F = 1$:

$$g_F = 2 \cdot \frac{2 + \frac{3}{4} - \frac{15}{4}}{2 \cdot 2} = 2 \cdot \frac{-1}{4}.$$

$$g_{F=1} = -\frac{1}{2}.$$

$F = 2$:

$$g_F = 2 \cdot \frac{6 + \frac{3}{4} - \frac{15}{4}}{2 \cdot 6} = 2 \cdot \frac{3}{12}.$$

$$g_{F=2} = +\frac{1}{2}.$$

(d) **Seven RF frequencies between $F = 1$ and $F = 2$**

Energy of a hyperfine sublevel (ΔF ignored as a constant offset between manifolds):

$$E(F, M_F) = E_0(F) + g_F \mu_B B_{\text{ext}} M_F.$$

RF transition frequency from $(F = 1, M)$ to $(F = 2, M')$:

$$h\nu = h\nu_0 + \mu_B B_{\text{ext}} (g_2 M' - g_1 M).$$

Substitute $g_1 = -\frac{1}{2}$, $g_2 = +\frac{1}{2}$:

$$h\nu = h\nu_0 + \frac{\mu_B B_{\text{ext}}}{2} (M' + M).$$

Magnetic-dipole selection: $\Delta F = \pm 1$, $\Delta M_F = 0, \pm 1$. Tabulate $(M + M')$:

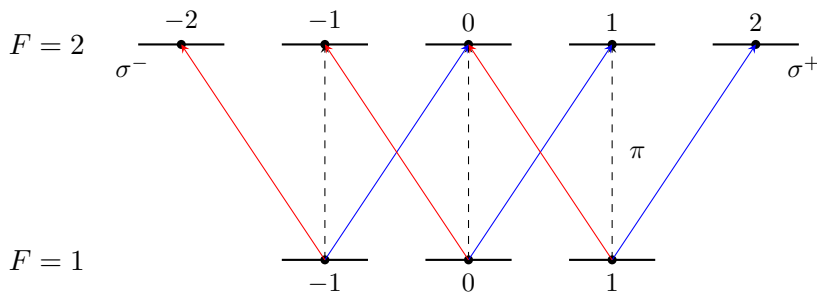
	$M = -1$	$M = 0$	$M = +1$
$M' = M - 1 (\sigma^-)$	–	–1	+1
$M' = M (\pi)$	–2	0	+2
$M' = M + 1 (\sigma^+)$	–1	+1	–

(missing entries: $M' = -2$ requires $M = -1 \rightarrow M' = -2$, giving $M + M' = -3$; $M' = +2$ requires $M = +1 \rightarrow M' = +2$, giving $M + M' = +3$).

Distinct values of $M + M'$:

$$\{-3, -2, -1, 0, +1, +2, +3\} \Rightarrow \boxed{\text{seven frequencies.}}$$

Longitudinal observation. Along $\mathbf{B}$, only $\sigma^\pm$ ($\Delta M_F = \pm 1$) radiation is emitted; π ($\Delta M_F = 0$) is forbidden. Visible shifts $(M + M') \in \{-3, -1, +1, +3\}$ from σ transitions only — four lines.



Solid (blue/red): $\sigma^\pm$ visible along $\mathbf{B}$. Dashed: π , not visible along $\mathbf{B}$.

Question 3 — ^{27}AlH band system, P/Q/R branches

Problem. Identify the terms in the rovibronic energy formula, give Λ, S for $X^1\Sigma^+$ and $A^1\Pi$, derive P, Q, R-branch frequencies, locate the band head and extract ν_{00} , $B'/(hc)$, equilibrium internuclear distances.

(a) Identifying the energy terms

$$E = E_e(R_0) + (n + \frac{1}{2})h\nu_v + B K(K + 1).$$

$E_e(R_0)$: electronic (Born–Oppenheimer) energy at the equilibrium internuclear distance R_0 .

$(n + \frac{1}{2})h\nu_v$: vibrational energy of small-amplitude harmonic motion about R_0 , $\nu_v = \frac{1}{2\pi} \sqrt{k/\mu}$, μ reduced nuclear mass.

$B K(K + 1)$: rigid-rotor energy at fixed R_0 , $B = \hbar^2/(2\mu R_0^2)$, K the total nuclear-rotation quantum number.

(b) Quantum numbers of $X^1\Sigma^+$ and $A^1\Pi$

Term symbol $^{2S+1}\Lambda$ with $\Lambda = |L_z|$ (electronic orbital projection on the internuclear axis):

$$\boxed{X^1\Sigma^+ : S = 0, \Lambda = 0; \quad A^1\Pi : S = 0, \Lambda = \hbar.}$$

(c) Branch frequencies

Transition energy (E' upper, E'' lower, $v' = v'' = 0$):

$$h\nu = h\nu_{00} + B'K'(K' + 1) - B''K''(K'' + 1).$$

P branch ($\Delta K = +1$ in emission, $K'' = K' + 1$, hence $K' = K'' - 1$):

Substitute $K' = K'' - 1$:

$$h\nu_P = h\nu_{00} + B'(K'' - 1)K'' - B''K''(K'' + 1).$$

Expand and collect powers of K'' :

$$h\nu_P = h\nu_{00} + B'(K''^2 - K'') - B''(K''^2 + K'').$$

Group:

$$\boxed{h\nu_P = h\nu_{00} - (B' + B'')K'' + (B' - B'')K''^2.} \quad (4)$$

Q branch ($K' = K''$):

$$h\nu_Q = h\nu_{00} + (B' - B'')K''(K'' + 1).$$

$$\boxed{h\nu_Q = h\nu_{00} + (B' - B'')K'' + (B' - B'')K''^2.} \quad (5)$$

R branch ($K' = K'' + 1$):

$$h\nu_R = h\nu_{00} + B'(K'' + 1)(K'' + 2) - B''K''(K'' + 1).$$

Expand:

$$h\nu_R = h\nu_{00} + B'(K''^2 + 3K'' + 2) - B''(K''^2 + K'').$$

Group:

$$\boxed{h\nu_R = h\nu_{00} + 2B' + (3B' - B'')K'' + (B' - B'')K''^2.} \quad (6)$$

(d) Band head, ν_{00} , $B'/(hc)$, equilibrium separations

Treat P and R together via Fortrat parameter m : $m = -K''$ (P, $m \leq -1$), $m = K'' + 1$ (R, $m \geq 1$).

Both (4) and (6) reduce to

$$h\nu(m) = h\nu_{00} + (B' + B'')m + (B' - B'')m^2.$$

Differentiate w.r.t. m :

$$\frac{d(h\nu)}{dm} = (B' + B'') + 2(B' - B'')m.$$

Set to zero:

$$m_{\text{head}} = \frac{B' + B''}{2(B'' - B')}.$$

For AlH the upper-state bond is longer, so $B' < B''$, giving $m_{\text{head}} > 0$: the head lies in the R branch.

Translate $m_{\text{head}} = K''_{\text{head}} + 1$:

$$\boxed{K''_{\text{head}} = \frac{B' + B''}{2(B'' - B')} - 1.} \quad (7)$$

Extracting $B'/(hc)$ from the Q branch. (5) written in wavenumbers ($\tilde{\nu} \equiv \nu/c$):

$$\tilde{\nu}_Q(K'') - \tilde{\nu}_{00} = (\tilde{B}' - \tilde{B}'')K''(K'' + 1).$$

Read from the figure: $\tilde{\nu}_Q(K'' = 22) \simeq 23\,200\text{ cm}^{-1}$, and the $K'' \rightarrow 0$ extrapolation of the Q branch gives $\tilde{\nu}_{00} \simeq 23\,480\text{ cm}^{-1}$ (this is also where Q meets the gap between R and P at low K''). Use $K''(K'' + 1) = 506$:

$$\tilde{B}' - \tilde{B}'' = \frac{23\,200 - 23\,480}{506}.$$

Numerator:

$$23\,200 - 23\,480 = -280.$$

Divide:

$$\tilde{B}' - \tilde{B}'' = \frac{-280}{506} = -0.553\text{ cm}^{-1}.$$

Add $\tilde{B}'' = 6.3907\text{ cm}^{-1}$:

$$\boxed{\tilde{B}' \equiv B'/(hc) = 5.84\text{ cm}^{-1}, \quad \tilde{\nu}_{00} \simeq 23\,480\text{ cm}^{-1}.}$$

Consistency check via R-branch head. (7):

$$K''_{\text{head}} = \frac{5.84 + 6.39}{2(6.39 - 5.84)} - 1 = \frac{12.23}{1.10} - 1.$$

Evaluate:

$$K''_{\text{head}} \simeq 11.1 - 1 = 10.1.$$

Head wavenumber from (6) at $m_{\text{head}} = 11.1$:

$$\tilde{\nu}_{\text{head}} = \tilde{\nu}_{00} + (\tilde{B}' + \tilde{B}'')m_{\text{head}} + (\tilde{B}' - \tilde{B}'')m_{\text{head}}^2.$$

Insert numbers:

$$\tilde{\nu}_{\text{head}} = 23\,480 + 12.23 \cdot 11.1 - 0.553 \cdot 123.2.$$

$$\tilde{\nu}_{\text{head}} = 23\,480 + 135.8 - 68.1 \simeq 23\,548\text{ cm}^{-1},$$

in agreement with the pile-up of R-branch points near $23\,545\text{ cm}^{-1}$ in the figure.

Equilibrium internuclear distances. For ^{27}AlH :

$$\mu = \frac{27 \cdot 1}{28}\text{ u} = 0.9643\text{ u} = 1.6014 \times 10^{-27}\text{ kg}.$$

$B = \hbar^2/(2\mu R_0^2) \Rightarrow B/(hc) = \hbar/(4\pi c\mu R_0^2)$, hence

$$R_0^2 = \frac{\hbar}{4\pi c\mu (B/(hc))}.$$

Numerator:

$$\hbar = 1.0546 \times 10^{-34}\text{ J s}.$$

Denominator pieces:

$$4\pi c = 4\pi \cdot 2.998 \times 10^8 = 3.767 \times 10^9\text{ m/s}.$$

$$4\pi c\mu = 3.767 \times 10^9 \cdot 1.6014 \times 10^{-27} = 6.033 \times 10^{-18}\text{ kg m/s}.$$

Constant prefactor:

$$\frac{\hbar}{4\pi c\mu} = \frac{1.0546 \times 10^{-34}}{6.033 \times 10^{-18}} = 1.748 \times 10^{-17}\text{ m}^2\text{ m}^{-1}.$$

Convert $\tilde{B}'' = 6.3907 \text{ cm}^{-1} = 638.07 \text{ m}^{-1}$:

$$R_0''^2 = \frac{1.748 \times 10^{-17}}{638.07} = 2.740 \times 10^{-20} \text{ m}^2.$$

$$R_0'' = 1.655 \times 10^{-10} \text{ m}.$$

$$\boxed{R_0(X^1\Sigma^+) = 1.66 \text{ \AA}.}$$

Convert $\tilde{B}' = 5.84 \text{ cm}^{-1} = 584 \text{ m}^{-1}$:

$$R_0'^2 = \frac{1.748 \times 10^{-17}}{584} = 2.993 \times 10^{-20} \text{ m}^2.$$

$$R_0' = 1.730 \times 10^{-10} \text{ m}.$$

$$\boxed{R_0(A^1\Pi) = 1.73 \text{ \AA}.}$$

The bond is longer in the electronically excited $A^1\Pi$ state, consistent with the smaller B' inferred from the Q-branch slope and the appearance of the R-branch head.

Question 4 — X-ray transitions and K-edge filtering

Problem. Interpret the Moseley-type formula and the screening constants σ_n, σ_m . Define K_α and the K-edge; find σ_1, σ_2 for Ca. Derive the filter inequality and find the highest- Z element whose K_α is absorbed by element $Z - 1$.

(a) Form of the formula and the meaning of $\sigma_{n,m}$

The hydrogenic Bohr formula for an electron in shell n of nuclear charge Z is $E_n = -hcR_\infty Z^2/n^2$. In a multi-electron atom an inner-shell electron sees the nuclear charge partially screened by the other electrons:

$$Z \rightarrow Z_{\text{eff}}^{(n)} = Z - \sigma_n.$$

The transition energy between two such shells is the difference of effective hydrogenic levels:

$$E_{nm} = hcR_\infty \left[\frac{(Z - \sigma_n)^2}{n^2} - \frac{(Z - \sigma_m)^2}{m^2} \right]. \quad (8)$$

σ_n : effective screening of the nucleus by all other electrons, as seen by an electron in shell n during the transition. It is approximately equal to the number of electrons whose mean radius is smaller than that of shell n , but is reduced by exchange/correlation and includes a partial contribution from electrons in the same shell. σ_n depends on n but only weakly on Z (Moseley empirical rule).

(b) K_α and the K-edge; σ for Ca

K_α : radiative transition $L \rightarrow K$ following the production of a 1s vacancy ($n = 2 \rightarrow n = 1, \Delta l = \pm 1$). Using (8) with $n = 1, m = 2$:

$$E_{K_\alpha} = hcR_\infty \left[(Z - \sigma_1)^2 - \frac{1}{4}(Z - \sigma_2)^2 \right].$$

K-edge: minimum photon energy capable of ejecting a 1s electron to the continuum; the K electron's binding energy. From (8) with $m \rightarrow \infty$:

$$E_K = hcR_\infty (Z - \sigma_1)^2.$$

Calcium (Z=20). Use $hcR_\infty = 13.606$ eV.

K-edge:

$$(Z - \sigma_1)^2 = \frac{4039}{13.606}.$$

Divide:

$$(Z - \sigma_1)^2 = 296.85.$$

Square root:

$$Z - \sigma_1 = 17.229.$$

$$\sigma_1 = 20 - 17.23 = 2.77.$$

K_α :

$$\frac{E_{K_\alpha}}{hcR_\infty} = 296.85 - \frac{1}{4}(20 - \sigma_2)^2.$$

Insert $E_{K_\alpha}/(hcR_\infty) = 3692/13.606 = 271.41$:

$$\frac{1}{4}(20 - \sigma_2)^2 = 296.85 - 271.41 = 25.44.$$

Multiply by 4:

$$(20 - \sigma_2)^2 = 101.74.$$

Square root:

$$20 - \sigma_2 = 10.087.$$

$$\sigma_2 = 9.91.$$

Comment: $\sigma_1 \simeq 2.77$ exceeds the naive value 1 (one other K electron) because the operative screening for the K_α /K-edge process includes contributions from outer electrons reorganising around the inner-shell hole. $\sigma_2 \simeq 9.91$ is approximately the count of electrons inside the L shell plus a substantial fraction of the L electrons themselves — reflecting that L electrons screen each other strongly in addition to the K shell.

(c) Filter inequality

Efficient absorption requires the K_α photon of element Z to lie above the K-edge of element $Z - 1$:

$$E_{K_\alpha}(Z) > E_{K\text{-edge}}(Z - 1).$$

Cancel hcR_∞ :

$$(Z - \sigma_1)^2 - \frac{1}{4}(Z - \sigma_2)^2 > (Z - 1 - \sigma_1)^2.$$

Let $a = Z - \sigma_1$, $b = Z - \sigma_2$:

$$a^2 - \frac{1}{4}b^2 > (a - 1)^2.$$

Expand the RHS:

$$a^2 - \frac{1}{4}b^2 > a^2 - 2a + 1.$$

Cancel a^2 :

$$-\frac{1}{4}b^2 > -2a + 1.$$

Multiply by -4 (flip inequality):

$$b^2 < 8a - 4.$$

Substitute back a, b :

$$(Z - \sigma_2)^2 < 8(Z - \sigma_1) - 4.$$

Expand the LHS and rearrange:

$$Z^2 - (2\sigma_2 + 8)Z + (\sigma_2^2 + 8\sigma_1 + 4) < 0.$$

Solve the quadratic; take the larger root for the upper bound:

$$Z < \frac{(2\sigma_2 + 8) + \sqrt{(2\sigma_2 + 8)^2 - 4(\sigma_2^2 + 8\sigma_1 + 4)}}{2}.$$

Simplify by factoring 4 inside the root:

$$\boxed{Z < \sigma_2 + 4 + \sqrt{(\sigma_2 + 4)^2 - (\sigma_2^2 + 8\sigma_1 + 4)}}. \quad (9)$$

(d) Highest Z' and energy mismatch

Insert $\sigma_1 = 2.77$, $\sigma_2 = 9.91$.

Outer term:

$$\sigma_2 + 4 = 13.91.$$

Inside the root:

$$(\sigma_2 + 4)^2 - (\sigma_2^2 + 8\sigma_1 + 4).$$

Compute pieces:

$$(\sigma_2 + 4)^2 = 13.91^2 = 193.49,$$

$$\sigma_2^2 = 9.91^2 = 98.21, \quad 8\sigma_1 = 22.16,$$

$$\sigma_2^2 + 8\sigma_1 + 4 = 98.21 + 22.16 + 4 = 124.37.$$

Subtract:

$$193.49 - 124.37 = 69.12.$$

Square root:

$$\sqrt{69.12} = 8.314.$$

Total:

$$Z < 13.91 + 8.31 = 22.22.$$

Hence

$$\boxed{Z'_{\max} = 22 \text{ (Ti, filtered by Sc, } Z' - 1 = 21\text{)}}.$$

Energy mismatch $\Delta E = E_{K_\alpha}(\text{Ti}) - E_{K\text{-edge}}(\text{Sc})$.

K-edge of Sc:

$$E_{K\text{-edge}}(21) = 13.606 (21 - 2.77)^2.$$

Compute:

$$(21 - 2.77)^2 = 18.23^2 = 332.33.$$

$$E_{K\text{-edge}}(21) = 13.606 \cdot 332.33 = 4521 \text{ eV}.$$

K_α of Ti:

$$E_{K_\alpha}(22) = 13.606 \left[(22 - 2.77)^2 - \frac{1}{4}(22 - 9.91)^2 \right].$$

Pieces:

$$(22 - 2.77)^2 = 19.23^2 = 369.79,$$

$$(22 - 9.91)^2 = 12.09^2 = 146.17, \quad \frac{1}{4} \cdot 146.17 = 36.54.$$

Subtract:

$$369.79 - 36.54 = 333.25.$$

Multiply:

$$E_{K_\alpha}(22) = 13.606 \cdot 333.25 = 4534 \text{ eV}.$$

Difference:

$$\Delta E = E_{K_\alpha}(\text{Ti}) - E_{K\text{-edge}}(\text{Sc}) \simeq 4534 - 4521 = 13 \text{ eV}.$$

The K_α of Ti lies only ~ 13 eV above the K-edge of Sc; for elements with $Z \geq 23$ (V), the K_α photon falls below the K-edge of $Z - 1$ and the simple “ Z filtered by $Z - 1$ ” rule fails.